Homework 2. solution

1. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha|t|}$ to be short-range or long range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty}(e^{-\alpha})^k = \sum_{k=1}^{\infty} q^k$ where $q = e^{-\alpha}$. The last sum is a geometric progression, therefore $1 + \sum_{k=1}^{\infty} q^k = \frac{1}{1-q}$. Thus, the sum is finite, that means that $R(t) = e^{-\alpha|t|}$ is short-range dependent.

2. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha t^2}$ to be short-range or long range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} e^{-\alpha k^2} \leq \sum_{k=1}^{\infty} e^{-\alpha k} = \sum_{k=1}^{\infty}(e^{-\alpha})^k = \sum_{k=1}^{\infty} q^k$ where $q = e^{-\alpha}$. The last sum is a geometric progression, therefore $1 + \sum_{k=1}^{\infty} q^k = \frac{1}{1-q}$. Thus, the sum is finite, that means that $R(t) = e^{-\alpha|t|}$ is short-range dependent.

3. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = e^{-\alpha|t|^{1.6}}$ to be short-range or long range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} e^{-\alpha k^{1.6}} \leq \sum_{k=1}^{\infty} e^{-\alpha k} = \sum_{k=1}^{\infty}(e^{-\alpha})^k = \sum_{k=1}^{\infty} q^k$ where $q = e^{-\alpha}$. The last sum is a geometric progression, therefore $1 + \sum_{k=1}^{\infty} q^k = \frac{1}{1-q}$. Thus, the sum is finite, that means that $R(t) = e^{-\alpha|t|^{1.6}}$ is short-range dependent.

4. Let $\alpha > 0$. Identify ranges of α for the covariance function $R(t) = 1/(1+|t|)^{\alpha}$ to be short-range or long range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} \frac{1}{(1+k)^{\alpha}}$ converges/diverges if and only if $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ converges/diverges because the ratio of the k-th summand of the first sum to the k-th summand of the second sum tends to 1 as $k \to \infty$, i.e. $\lim_{k\to\infty} \frac{1}{(1+k)^{\alpha}} \Big/ [\frac{1}{k^{\alpha}}] = 1$. From the theory on convergence of series, we know $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ converges if $\alpha > 1$ and divergence otherwise. Thus, the covariance function $R(t) = 1/(1+|t|)^{\alpha}$ is short-range dependent if $\alpha > 1$ and long range dependent if $0 < \alpha \leq 1$.

5. Identify if the covariance function $R(t) = 1/(1+t^2)$ is short-range or long-range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} \frac{1}{1+k^2}$ converges/diverges if and only if $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges/diverges because the ratio of the k-th summand of the first sum to the k-th summand of the second sum tends to 1 as $k \to \infty$, i.e. $\lim_{k\to\infty} \frac{1}{1+k^2} \Big/ [\frac{1}{k^2}] = \lim_k \frac{k^2}{1+k^2} = \lim_k \frac{1}{1/k^2+1} = 1$. From the theory on convergence of series, we know $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ converges if $\alpha > 1$. In our case, $\alpha = 2$ which is larger than 1. Thus, the covariance function $R(t) = 1/(1+t^2)$ is short-range dependent.

6. Identify if the covariance function $R(t) = 1/(1+\sqrt{|t|})^2$ is short-range or long-range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} 1/(1+\sqrt{k})^2$ converges/diverges if and only if $\sum_{k=1}^{\infty} \frac{1}{k}$ converges/diverges because the ratio of the k-th summand of the first sum to the k-th summand of the second sum tends to 1 as $k \to \infty$, i.e. $\lim_{k\to\infty} \frac{1}{(1+\sqrt{k})^2} \Big/ [\frac{1}{k}] = \lim_k \frac{k}{1+2\sqrt{k}+k} = \lim_k \frac{1}{1/k+2/\sqrt{k}+1} = 1$. From the theory on convergence of series, we know $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ diverges if $0 < \alpha \leq 1$. In our case, $\alpha = 1$. Thus, the covariance function $R(t) = 1/(1+\sqrt{|t|})^2$ is long-range dependent.

7. Identify if the covariance function $R(t) = 4/(5+5\sqrt{|t|})$ is short-range or long-range dependent.

   Note that $\sum_{k=1}^{\infty} R(k) = \sum_{k=1}^{\infty} 4/(5+5\sqrt{|k|})$ converges/diverges if and only if $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ converges/diverges because the ratio of the k-th summand of the first sum to the k-th summand of the second sum tends to a nonzero constant as $k \to \infty$, i.e. $\lim_{k\to\infty} \frac{4}{5+5\sqrt{k}} \Big/ [\frac{1}{\sqrt{k}}] = \lim_k \frac{4\sqrt{k}}{5+5\sqrt{k}} = \lim_k \frac{4}{4/\sqrt{k}+5} = 4/5$. From the theory on convergence of series, we know $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ diverges if $0 < \alpha \leq 1$. In our case, $\alpha = 1/2$ which is smaller than 1. Thus, the covariance function $R(t) = 4/(5+5\sqrt{|t|})$ is long-range dependent.